Homework 2 – k Sample Inference

1 Question 1: Why do we Correct for Multiple Comparisons?

A student asked me what the point of correcting for multiple comparisons was. They made a two interesting points:

1. They are different outcomes, so, unlike the coin flip example (where the coin is the same) it doesn't makes sense to combine them.
2. If a manuscript presented three studies with t-tests, they don't correct for multiple comparisons across *those* tests.

When we conduct all pairwise comparisons after an omnibus test like ANOVA, we are testing a single family of hypotheses to answer one overarching question ("Which of these groups are different?"). Every additional test provides another opportunity to commit a Type I error from random noise. Similar to the coin, the probability of making at least one false error across J independent tests grows exponentially: $1-(1-\alpha)^J$. Note that our pairwise comparisons are not independent, so it's slightly different.

We don't correct for inferences across separate studies because they typically test distinct hypotheses using independent samples. Our attempt to address the multiple comparisons problem is specifically confined to the issue of increased Type I error within a single decision-making context. If we don't, we can find ourselves in a p-hacking-like situation, where an investigator might search through many tests to find one $p < 0.05$ (a potential false discovery), rather than making decisions to best position themselves for finding a true discovery.

Here, we will quickly demonstrate why we correct for the multiple comparisons problem when conducting several tests to answer the same question.

1.1 Part a

Complete the following steps in $R = 1000$ simulations.

1. Simulate data for $k = 5$ samples $n = 30$ drawn from a standard Gaussian distribution.
2. Use `pairwise.t.test(..., p.adjust.method="none")` to compute unadjusted pairwise t-tests and record if any comparison yields statistically discernible support for a non-zero mean.
3. Use `rstatix::tukey_hsd(...)` to compute the Tukey's HSD adjusted pairwise tests and record if any comparison yields statistically discernible support for a non-zero mean.

In the end, describe the Type I error rates.

Solution

Because data for all groups are drawn from the same distribution, there's no mean difference between 5 groups (the null hypothesis is true).

If the 10 pairwise comparisons were independent, we would expect the probability of making at least 1 Type I error (FWER) to be approximately 0.4.

$$P(\text{at least 1}) = 1 - P(\text{none}) = 1 - (1-\alpha)^{10} = 1 - (1-0.05)^{10} = 0.40$$

The Type I error rate in the unadjusted t-tests is 0.286, still lower than 0.4. This is because the pairwise comparisons are not independent (e.g., comparison between group 1 and 2 and comparison between group 1 and 3 both use data from group 1).

The unadjusted t-tests also yield a Type I error rate 5 times higher than in the Tukey's HSD adjusted pairwise tests (0.054). This means that, if we don't correct for the multiple comparison problem, we're much more likely to make Type I error when performing the same statistical test on the same dataset many times.

```
set.seed(7272)

reject1 = rep(NA, 1000)
reject2 = rep(NA, 1000)

for(i in 1:1000){
  sim = data.frame(
    g1 = rnorm (30, mean = 0, sd = 1),
    g2 = rnorm (30, mean = 0, sd = 1),
    g3 = rnorm (30, mean = 0, sd = 1),
    g4 = rnorm (30, mean = 0, sd = 1),
    g5 = rnorm (30, mean = 0, sd = 1))
  sim = sim |>
    pivot_longer(cols = everything(), names_to = "Group", values_to = "Value")

  unadjusted = pairwise.t.test(sim$Value, sim$Group, p.adjust.method = "none")
  reject1[i] = any(unadjusted$p.value < 0.05, na.rm = TRUE)
  # reject1 will be a vector of TRUE/FALSE
  hsd = rstatix::tukey_hsd(sim, formula = Value ~ Group)
  reject2[i] = any(hsd$p.adj < 0.05, na.rm = TRUE)
}

mean(reject1)
```

```
[1] 0.286
```

```
mean(reject2)
```

```
[1] 0.054
```

1.2 Part b

Redo the simulation, changing to adjust according to the false discovery rate `p.adjust.method="BH"` and comment on the differences.

Solution

In a world where the null is true, FDR ($p = 0.048$) is similar to FWER calculated in Part a ($p = 0.054$).

```
set.seed(7272)

reject3 = rep(NA, 1000)

for(i in 1:1000){
  sim = data.frame(
    g1 = rnorm (30, mean = 0, sd = 1),
    g2 = rnorm (30, mean = 0, sd = 1),
    g3 = rnorm (30, mean = 0, sd = 1),
    g4 = rnorm (30, mean = 0, sd = 1),
    g5 = rnorm (30, mean = 0, sd = 1))
  sim = sim |>
    pivot_longer(cols = everything(), names_to = "Group", values_to = "Value")

  BH = pairwise.t.test(sim$Value, sim$Group, p.adjust.method = "BH")
  reject3[i] = any(BH$p.value < 0.05, na.rm = TRUE)
}

mean(reject3)
```

```
[1] 0.048
```

2 Question 2: Power Analyses

Revisit the MFAP4 study from class (Bracht et al. 2016).

2.1 Part a

Assume the researchers were targeting a moderate effect size of Cohen's $f = 0.25$, setting significance level $\alpha = 0.05$ and desired power $1 - \beta = 0.80$. Were they adequately powered?

Note: The actual effect size is $f = 0.55$, but researchers usually do not know that ahead of time and defaulted to moderate, instead of specifying a large effect.

What is the required sample size per group (n) and the overall total sample size (N)? Write the `R` code using the `pwr` package to calculate this sample size. This package assumes all samples are of the same size so $N = nk$, where k is the number of samples. Note that the best answers will have a corresponding graphical display.

Solution

In order to detect a moderate effect size of $f = 0.25$ with 80% power, we need 40 observations per group and a total of 200 observations for 5 groups. The study was adequately powered because its sample was 97 + 176 + 135 + 67 + 67 = 542, well above 200.

```
pwr_analysis = pwr.anova.test(k = 5,
                              f = 0.25,
                              sig.level = 0.05,
                              power = 0.80)
ceiling(pwr_analysis$n)
```

```
[1] 40
```

```
f = 0.25
k = 5
alpha = 0.05
n = 40
N = n * k

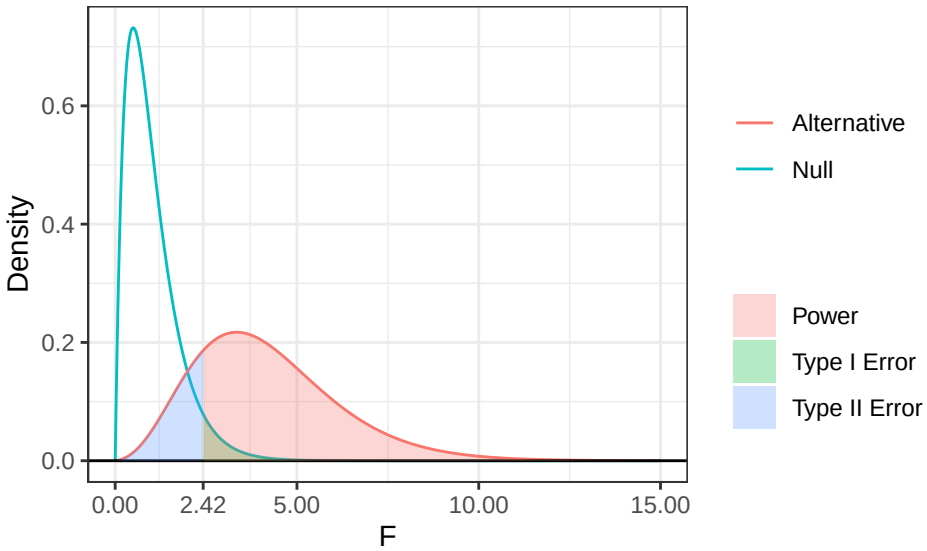
df1 = k - 1
df2 = N - k
lambda = f^2 * N

F_crit <- qf(1 - alpha, df1, df2)
```

```
ggdat = tibble(t = seq(0, 15, length.out = 1000)) |>
  mutate(pdf.h0 = df(x = t, df1 = df1, df2 = df2),
         pdf.ha = df(x = t, df1 = df1, df2 = df2, ncp = lambda))

ggplot(ggdat) +
  geom_line(aes(x = t, y = pdf.h0, color = "Null")) +
  geom_line(aes(x = t, y = pdf.ha, color = "Alternative")) +
  geom_hline(yintercept = 0) +
  geom_ribbon(data = ggdat |> filter(t >= F_crit),
              aes(x = t, ymin = 0, ymax = pdf.h0, fill = "Type I Error"), alpha = 0.3) +
  geom_ribbon(data = ggdat |> filter(t < F_crit),
              aes(x = t, ymin = 0, ymax = pdf.ha, fill = "Type II Error"), alpha = 0.3) +
  geom_ribbon(data = ggdat |> filter(t >= F_crit),
              aes(x = t, ymin = 0, ymax = pdf.ha, fill = "Power"), alpha = 0.3) +
  theme_bw() +
  scale_x_continuous("F", breaks = c(0, round(F_crit, 2), seq(5, 15, by = 5))) +
  # round F_crit to 2 decimals
  labs(y = "Density", color = "", fill = "")
```

Figure 1: Moderate (f = 0.25, k = 5); n per group = 40 (N = 200)



2.2 Part b

The project was funded by several agencies. Suppose that a funding agency suggested funding limiting the researchers to $N = 150$. Evaluate this suggestion in the context of the ANOVA. Would that be large enough to be adequately powered?

Compute the non-centrality parameter

$$\lambda = k \cdot n \cdot f^2$$

and use the non-central F distribution to compute the power under H_a for Cohen's $f = 0.25$. Calculate the statistical power achieved under this budget constraint. Note that the best answers will have a corresponding graphical display.

Solution

The total sample size is not enough for the sample to be adequately powered at 80%. Each group has a sample size of 30. The statistical power achieved under this budget constraint is 0.67.

The non-centrality parameter is $\lambda = 9.375$ and is used to construct the F-distribution under H_a.

```
(pwr_analysis = pwr.anova.test(k = 5,
                               f = 0.25,
                               n = 150 / 5,
                               sig.level = 0.05))
```

```
     Balanced one-way analysis of variance power calculation

              k = 5
              n = 30
              f = 0.25
      sig.level = 0.05
          power = 0.6675964

NOTE: n is number in each group
```

```
# whichever parameters left empty, the function will calculate
```

```
f = 0.25
k = 5
alpha = 0.05
n = 30
N = n * k

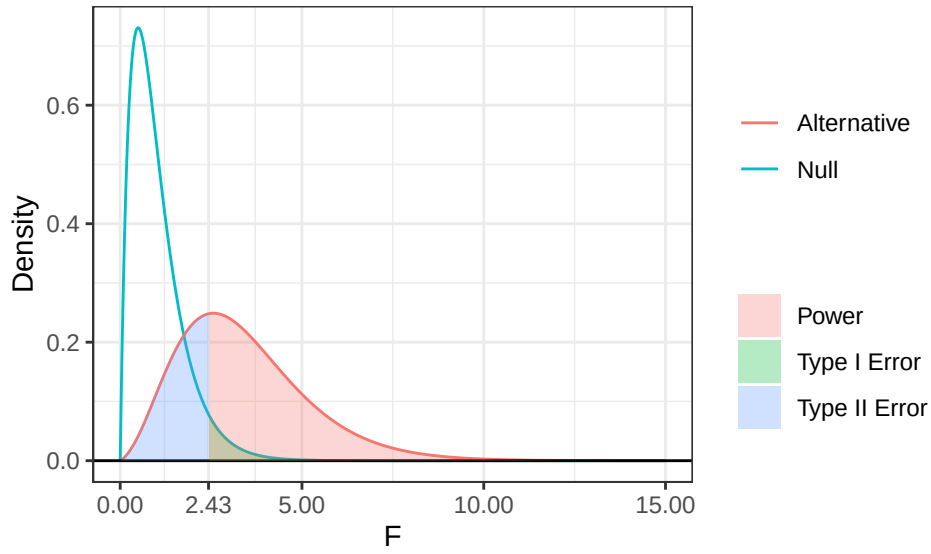
df1 = k - 1
df2 = N - k
lambda = f^2 * N

F_crit = qf(1 - alpha, df1, df2)
```

```
ggdat = tibble(t = seq(0, 15, length.out = 1000)) |>
  mutate(pdf.h0 = df(x = t, df1 = df1, df2 = df2),
         pdf.ha = df(x = t, df1 = df1, df2 = df2, ncp = lambda))

ggplot(ggdat) +
  geom_line(aes(x = t, y = pdf.h0, color = "Null")) +
  geom_line(aes(x = t, y = pdf.ha, color = "Alternative")) +
  geom_hline(yintercept = 0) +
  geom_ribbon(data = ggdat |> filter(t >= F_crit),
              aes(x = t, ymin = 0, ymax = pdf.h0, fill = "Type I Error"), alpha = 0.3) +
  geom_ribbon(data = ggdat |> filter(t < F_crit),
              aes(x = t, ymin = 0, ymax = pdf.ha, fill = "Type II Error"), alpha = 0.3) +
  geom_ribbon(data = ggdat |> filter(t >= F_crit),
              aes(x = t, ymin = 0, ymax = pdf.ha, fill = "Power"), alpha = 0.3) +
  theme_bw() +
  scale_x_continuous("F", breaks = c(0, round(F_crit, 2), seq(5, 15, by = 5))) +
  # round F_crit to 2 decimals
  labs(y = "Density", color = "", fill = "")
# less overlap, more power, more overlap, less power
```

Figure 2: Moderate (f = 0.25, k = 5); n per group = 30 (N = 150)



2.3 Part c

In the article the researchers state:

> *However, the statistical power was not sufficient to allow an accurate discrimination of no to moderate stages of hepatic fibrosis (F0–F2) from severe fibrosis and cirrhosis (F3 and F4).*

The European Association for the Study of the Liver (EASL) provides treatment recommendations based on these stages.

- Stages 0-1: Treatment is deferred with regularly assessment
- Stage 2: Treatment is justified for those in stage 2
- Stage 3-4: Treatment is a priority for those in stages

Suppose the researcher wanted to perform a follow-up test where they want to ensure adequate power not just for the overall ANOVA F-test, but specifically to detect a pairwise differences between stages 1 and 2. Explain why the sample size computed in Part a may not be enough and use `PMCMRplus::power.tukey.test()` to compute a sample size that attains adequate power.

Note: The researchers would have had to do this *a priori*. Because we aren't medical experts, you can use sample estimates using the original data.

$$
\begin{aligned}
n &= \text{answer from part a} \\
\mu_\Delta &= 0.25 \\
\sigma^2_{\text{pooled}} &= \frac{(176-1)0.85^2 + (135-1)0.84^2}{176+135-2} = 0.7152 \\
\sigma_{\text{within}} &= 0.77
\end{aligned}
$$

Solution

The sample size computed in Part a is not sufficient to detect a difference between stage 1 and 2 specifically, because:

- The computation in Part a uses Cohen's f, which reflects the spread of group means around within-group variability. So the power calculation using this effect size doesn't account for any particular pair's difference.
- The power analysis for omnibus ANOVA uses data from 5 groups, but the pairwise power analysis only uses information from 2 groups.

The sample size calculated in Part a only allows us to detect difference between stage 1 and 2 with 0.24 power. I use `PMCMRplus::power.tukey.test()` and `repeat` loop to compute a sample size that attains adequate power. We need 150 observations for each group to attain power of 0.80.

```
pwr12 = PMCMRplus::power.tukey.test(n = 40,
                                    groups = 2,
                                    delta = 0.25, #assumed mean population difference between stage 1 and 2
                                    within.var = 0.77,
                                    sig.level = 0.05)
(pwr12$power)
```

```
[1] 0.2421382
```

```
x = 40
repeat{
  pwr12 = PMCMRplus::power.tukey.test(n = x,
                                      groups = 2,
                                      delta = 0.25,
                                      within.var = 0.77^2, # square the SD
                                      sig.level = 0.05)
  if(pwr12$power >= 0.80) {
    # using == 0.80 will likely never be true, because power is continuous
    break
  }
  x = x+1
  # put at the end of loop, so that it's only updated after checked with condition
}
print(c(n = x, power = pwr12$power))
```

```
          n       power
150.0000000   0.8003124
```

2.4 Part d

How does the answer to Part c change if they wanted to have adequate power for *all* pairwise comparisons? Use `Superpower::ANOVA_design()` to set up the ANOVA set up.

Note: Use design "5b" and the following.

$$\mu = \{2.92, 3.11, 3.36, 4.04, 4.35\}$$

$$\sigma^2 = \frac{(97-1)0.87^2 + (176-1)0.85^2 + (135-1)0.84^2 + (67-1)1.00^2 + (67-1)0.90^2}{97+176+135+67+67-5} = 0.80$$

Compute the power of the researcher's design (part a) using `Superpower::ANOVA_power()`. If the researchers were able to oversample by double, a common approach to addressing this issue, would their power be at least 80% for all pariwise comparisons?

Note: This code can take a minute. You may want to set `cache: true` in your `R` chunk yaml header. You will also see more power because the actual effect size was 0.55.

Solution

`Superpower::ANOVA_power()`estimates power via Monte Carlo simulation. It simulates a random dataset that has the same number of groups, sample size per group, means and SDs as the researchers assumed/expected (noted that power analysis is often conducted a priori). It runs the ANOVA and pairwise tests on this dataset. Then, it calculates how many out of the total number of simulations (which we specify) a mean difference (between any group — H_a: at least 1 mean is different) is detected. Thus, if the differences that the researchers expect does exist, the test can detect them that proportion of the times.

The researcher's design in Part a yields 100% power to detect **at least a mean difference** among 5 groups, but the power to detect some pairwise comparisons are extremely low, namely the difference between stage 1 and 2 (16.5%), stage 2 and 3 (22.7%), and stage 4 and 5 (35.3%). Difference between stage 1 and 3 can only be detected with approximately 60% power.

Even when sample size for each group is doubled, the powers for all pairwise comparison are not altogether beyond 80% (e.g., the power to detect difference between stage 1 and 2 is 26%).

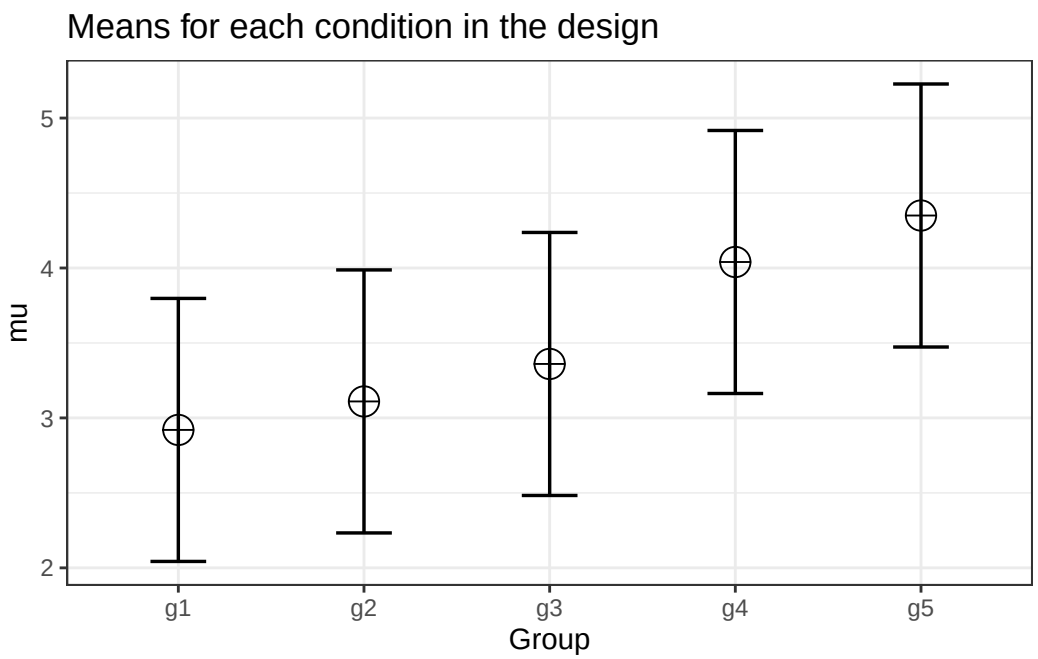
The researchers need about 330 observations per group in order to have adequate power for all pairwise comparisons.

```
# n = 40

means = c(2.92, 3.11, 3.36, 4.04, 4.35)
```

```r
var_pooled = ((97-1)*0.87^2 + (176-1)*0.85^2 + (135-1)*0.84^2 +
              (67-1)*1.00^2 + (67-1)*0.90^2) / (97+176+135+67+67-5)
sd_pooled = sqrt(var_pooled)

design_40 = ANOVA_design(design = "5b",
                         n = 40,
                         mu = means,
                         sd = sd_pooled,
                         labelnames = c("Group", "g1", "g2", "g3", "g4", "g5")
# first element is the name of the factor (IV) and the rests are names of
# each level/group
)
```



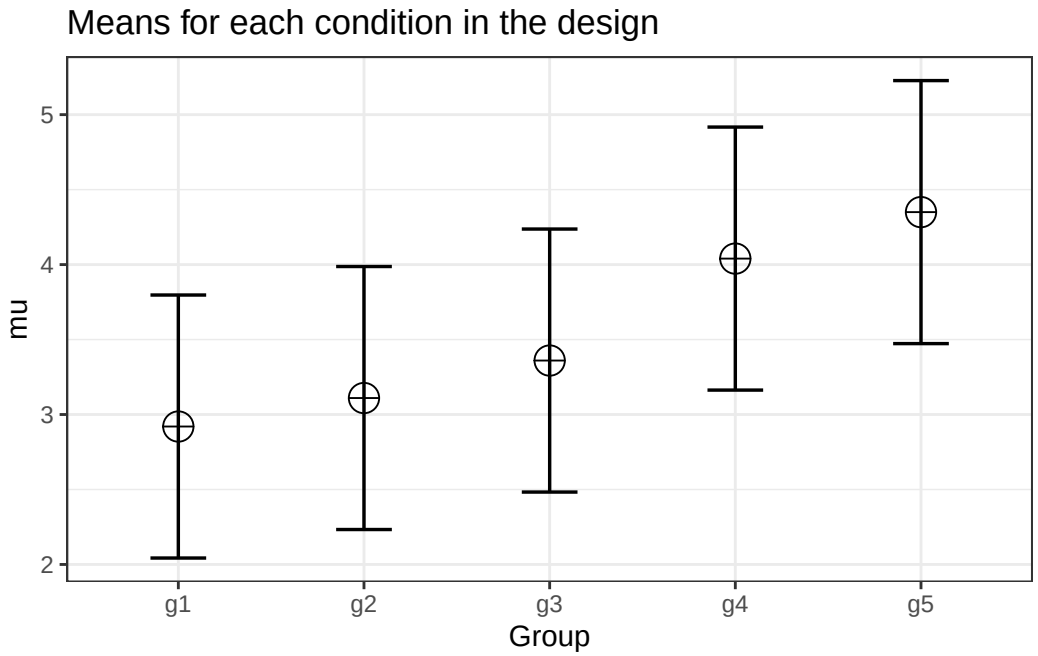

```r
# omnibus ANOVA
power_40 = ANOVA_power(design_40,
                       alpha_level = 0.05,
                       nsims = 1000,
                       seed = 7272)
```

```
Power and Effect sizes for ANOVA tests
            power effect_size
anova_Group   100      0.2943

Power and Effect sizes for pairwise comparisons (t-tests)
                    power effect_size
p_Group_g1_Group_g2  16.5      0.2264
p_Group_g1_Group_g3  60.4      0.5026
p_Group_g1_Group_g4 100.0      1.2831
p_Group_g1_Group_g5 100.0      1.6442
p_Group_g2_Group_g3  22.7      0.2786
p_Group_g2_Group_g4  99.8      1.0617
p_Group_g2_Group_g5  99.9      1.4246
p_Group_g3_Group_g4  94.0      0.7822
p_Group_g3_Group_g5  99.9      1.1453
p_Group_g4_Group_g5  35.3      0.3653
```

```r
# n = 80

design_80 = ANOVA_design(design = "5b",
                         n = 80,
                         mu = means,
                         sd = sd_pooled,
                         labelnames = c("Group", "g1", "g2", "g3", "g4", "g5")
)
```




```
power_80 = ANOVA_power(design_80,
                       alpha_level = 0.05,
                       nsims = 1000,
                       seed = 7272)
```

```
Power and Effect sizes for ANOVA tests
            power effect_size
anova_Group   100      0.2873

Power and Effect sizes for pairwise comparisons (t-tests)
                     power effect_size
p_Group_g1_Group_g2  26.3      0.2085
p_Group_g1_Group_g3  88.8      0.5005
p_Group_g1_Group_g4 100.0      1.2735
p_Group_g1_Group_g5 100.0      1.6293
p_Group_g2_Group_g3  43.7      0.2932
p_Group_g2_Group_g4 100.0      1.0683
p_Group_g2_Group_g5 100.0      1.4244
p_Group_g3_Group_g4  99.6      0.7731
p_Group_g3_Group_g5 100.0      1.1280
p_Group_g4_Group_g5  59.2      0.3555
```

```
x = 80

repeat{
  design = ANOVA_design(design = "5b",
                        n = x,
                        mu = means,
                        sd = sd_pooled,
                        labelnames = c("Group", "g1", "g2", "g3", "g4", "g5"))
  power = ANOVA_power(design,
                      alpha_level = 0.05,
                      nsims = 1000,
                      seed = 7272)

  if (all(power$pc_results$power >= 80)) {
    break
  }
  x = x+1
}
print(c(x, power$pc_results))
```

3 Question 3: An Application

Yang et al. (2023) investigate whether low-dose nicotine can counteract age-related metabolic decline. The proposed mechanism is through NAD^+, which is a chemical compound essential for metabolic energy production. The idea is that low-dose nicotine could help improve and/or maintain the recycling of NAD^+ by activating the salvage enzyme NAMPT via SIRT1. This could, in turn, restore NAD^+ homeostasis, which is the biological balance maintained within a cell between the production, recycling, and degradation of nicotinamide adenine dinucleotide (NAD^+).

Note: It is low-dose nicotine. Cigarette smoking delivers massive doses of nicotine along with other harmful chemicals. This is not an endorsement for smoking.

Male mice across distinct life stages – 2, 6, and 16 months – were categorized to evaluate natural, age-dependent metabolic and enzymatic shifts in two conditions.

- Treatment: low-dose nicotine (administered in drinking water)
- Control: no-dose placebo (administered in drinking water)

The researchers then aimed to compare treatment against control in the following samples:

- 2 months old, control
- 6 months old, control
- 16 months old, control
- 16 months old, treatment

Using PET imaging, the researchers measured age-induced inflammatory glucose hypermetabolism (SUV_{max}). They want to show that this quantity increases with age and those in the nicotine group saw levels comparable to younger mice.

Table 1: SUV_{max} measurements in the heart across age and treatment groups

Replicate	2M	6M	16M	16M, Treatment
1	1.48969	4.36000	7.30182	4.69304
2	1.30168	4.28807	7.12416	2.51326
3	1.30620	3.64184	8.28764	2.38019

3.1 Part a

State the null (H_0) and alternative (H_a) hypotheses for the omnibus test.

Solution

For the omnibus test,

H_0: all 4 groups have the same SUV_{max} measurements ($\mu_{2M} = \mu_{6M} = \mu_{16M} = \mu_{16MT}$)

H_a: at least 1 mean is different

3.2 Part b

Perform some exploratory data analysis to provide empirical information about the researchers question.

Solution

SUV_{max} seems to increase with age, and the nicotine treatments reduce SUV_{max} in the treated 16-month-old group. Variability differs across groups (e.g., there's a huge outlier in the treated 16-month-old group).

```
df = data.frame(
  ID = c(1, 2, 3),
  m2 = c(1.48969, 1.30168, 1.30620),
  m6 = c(4.36000, 4.28807, 3.64184),
  m16 = c(7.30182, 7.12416, 8.28764),
  m16t = c(4.69304, 2.51326, 2.38019))

df = df |>
  pivot_longer(cols = c(m2, m6, m16, m16t),
               names_to = "Group",
               values_to = "Value")

y_limits = range(df$Value)
```

```
summary = df |>
  mutate(Group = factor(Group, levels = c("m2", "m6", "m16", "m16t"))) |>
  group_by(Group) |>
  summarize(
    Mean = mean(Value),
    SD = sd(Value),
    Median = median(Value),
    IQR = IQR(Value),
```

```
    Min = min(Value),
    Max = max(Value)
  )
```

```
summary |> knitr::kable(digits = 2)
```

Table 2: Descriptive Statistics for SUVmax Across Age and Treatment Group

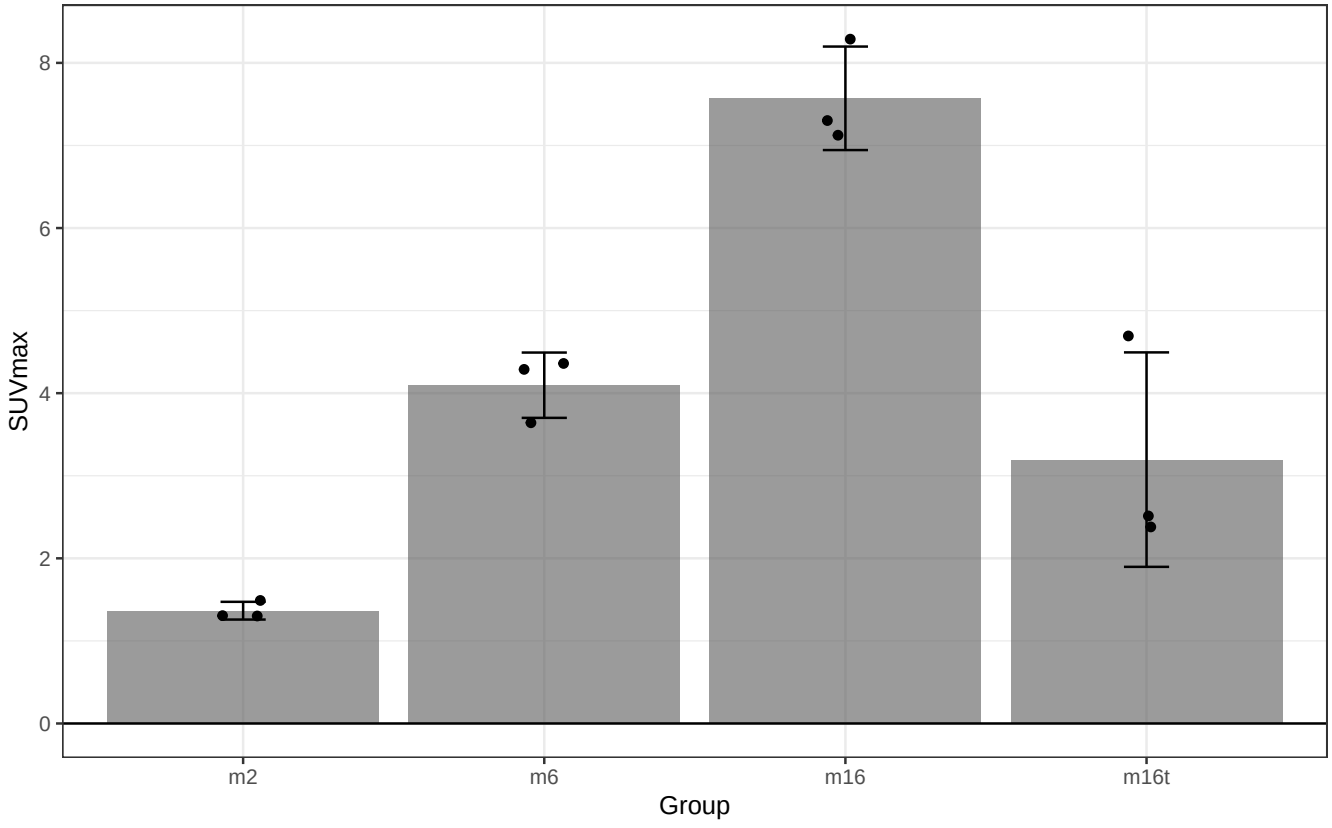
Group	Mean	SD	Median	IQR	Min	Max
m2	1.37	0.11	1.31	0.09	1.30	1.49
m6	4.10	0.40	4.29	0.36	3.64	4.36
m16	7.57	0.63	7.30	0.58	7.12	8.29
m16t	3.20	1.30	2.51	1.16	2.38	4.69

```
df = data.frame(
  ID = c(1, 2, 3),
  m2 = c(1.48969, 1.30168, 1.30620),
  m6 = c(4.36000, 4.28807, 3.64184),
  m16 = c(7.30182, 7.12416, 8.28764),
  m16t = c(4.69304, 2.51326, 2.38019))

df = df |>
  pivot_longer(cols = c(m2, m6, m16, m16t),
               names_to = "Group",
               values_to = "Value")
```

```
ggplot(summary) +
  geom_bar(aes(x = Group, y = Mean), stat = "identity", alpha = 0.6) +
  geom_errorbar(aes(x = Group, ymin = Mean - SD, ymax = Mean + SD), width = 0.15) +
  geom_point(data = df,
             aes(x = factor(Group, levels = c("m2", "m6", "m16", "m16t")), y = Value),
             position = position_jitter(width = 0.08, height = 0)) +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Group", y = "SUVmax")
```

Figure 3: SUVmax Across Age and Treatment Groups



3.3 Part c

Perform and report the results of the omnibus test. Make sure to evaluate the conditions for using the method you choose.

Solution

The variable of interest is quantitative; the sample is presumed to be representative of the population; the sample observations are presumed to be independent. From Figure 3 and Table 2, we observe that groups don't have equal variances, so we cannot perform ANOVA and have to use Welch's ANOVA. There's no reason to assume normality, but this test is generally robust to normality, as long as we fulfill other conditions of use.

There is a significant difference between groups regarding SUV_{max} ($F_{3,3.55} = 98.46, p < 0.001, \omega^2 = 0.97$). The small sample size technically increases Type II error (the probability of failing to reject H_0 when H_a is true), but in this case, we were still able to find statistically discernible support for H_a because the effect size is so large.

```
(welch_result = oneway.test(Value ~ Group, data = df, var.equal = FALSE))
```

```
    One-way analysis of means (not assuming equal variances)

data:  Value and Group
F = 98.464, num df = 3.0000, denom df = 3.5496, p-value = 0.0006851
```

```
(omega_squared(welch_result))
```

```
# Effect Size for ANOVA

Omega2 |       95% CI
----------------------
0.97   | [0.83, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

3.4 Part d

Conduct post-hoc pairwise comparisons to determine if age-induced inflammatory glucose hypermetabolism increase with age and whether low-dose nicotine reduces age-induced inflammatory glucose hypermetabolism.

```
games_howell_result = df |>
  games_howell_test(Value ~ Group)
# the estimate is calculated by group 2 - group 1
```

```
# Does hypermetabolism increase with age? (m2 > m6 > m16)
age = games_howell_result |>
  filter(group1 != "m16t" & group2 != "m16t") |>
  select(group1, group2, estimate, conf.low, conf.high, p.adj)
```

```
age |> knitr::kable(digits = 2)
```

Table 3: Difference in SUVmax Between Age Groups

group1	group2	estimate	conf.low	conf.high	p.adj
m16	m2	-6.21	-8.60	-3.81	0.01
m16	m6	-3.47	-5.39	-1.56	0.01
m2	m6	2.73	1.31	4.15	0.01

```
# Does low-dose nicotine reduce SUV max at 16 months? (m16 vs m16t)
nicotine = games_howell_result |>
  filter(group1 == "m16" & group2 == "m16t") |>
  select(group1, group2, estimate, conf.low, conf.high, p.adj)
```

```
nicotine |> knitr::kable(digits = 2)
```

Table 4: Difference in SUVmax Between Treatment and Control Groups

group1	group2	estimate	conf.low	conf.high	p.adj
m16	m16t	-4.38	-8.51	-0.24	0.04

```
label_map = c(m2 = "2 Months", m6 = "6 Months", m16 = "16 Months", m16t = "16 Months (T)")

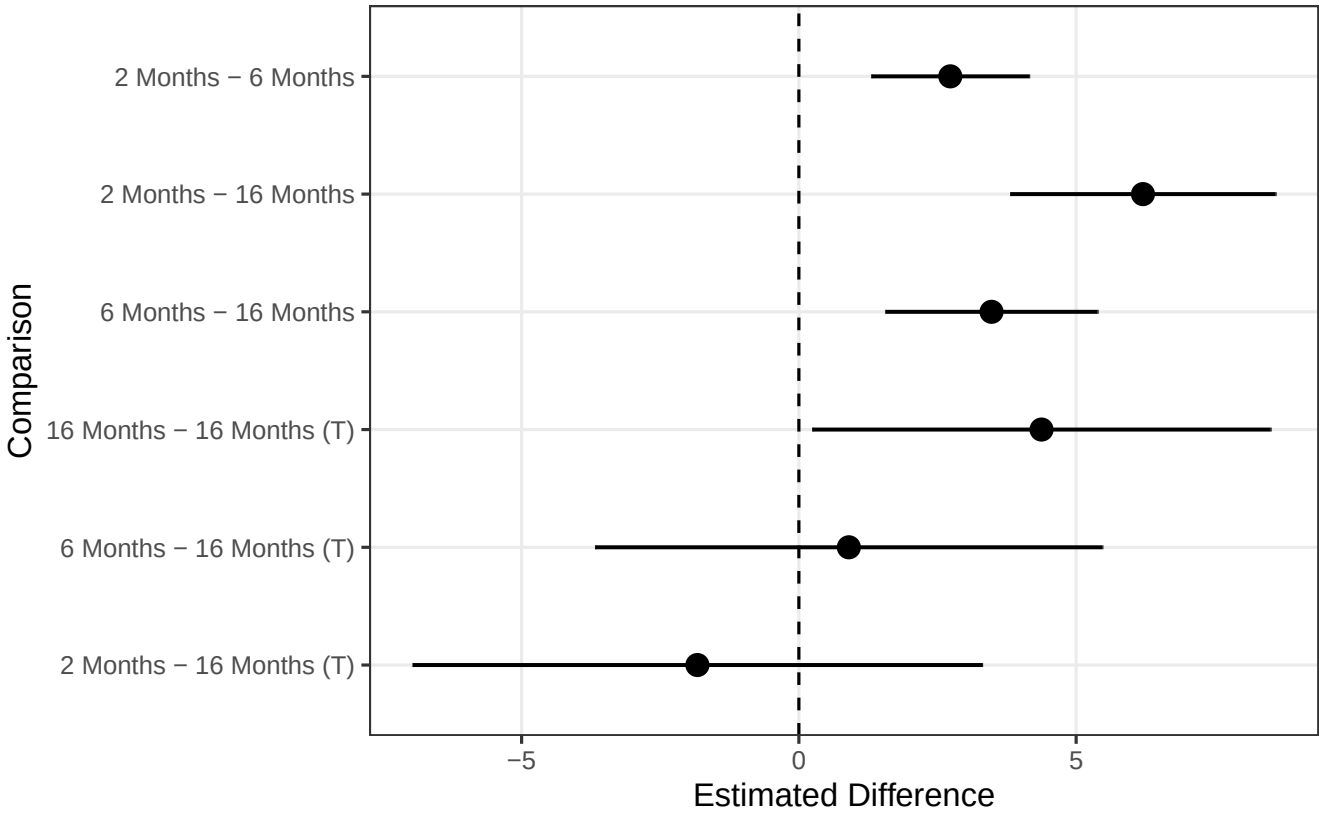
ggdat = tibble(
  comparison = c(
```

```
    paste(label_map["m2"], "-", label_map["m16t"]),
    paste(label_map["m6"], "-", label_map["m16t"]),
    paste(label_map["m16"], "-", label_map["m16t"]),
    paste(label_map["m6"], "-", label_map["m16"]),
    paste(label_map["m2"], "-", label_map["m16"]),
    paste(label_map["m2"], "-", label_map["m6"])
  ),
  estimate = c(
    games_howell_result$estimate[games_howell_result$group1 == "m16t" & games_howell_result$group2 == "m2"],
    games_howell_result$estimate[games_howell_result$group1 == "m16t" & games_howell_result$group2 == "m6"],
    -games_howell_result$estimate[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m16t"],
    -games_howell_result$estimate[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m6"],
    -games_howell_result$estimate[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m2"],
    games_howell_result$estimate[games_howell_result$group1 == "m2"   & games_howell_result$group2 == "m6"]
  ),
  conf.low = c(
    games_howell_result$conf.high[games_howell_result$group1 == "m16t" & games_howell_result$group2 == "m2"],
    games_howell_result$conf.high[games_howell_result$group1 == "m16t" & games_howell_result$group2 == "m6"],
    -games_howell_result$conf.high[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m16t"],
    -games_howell_result$conf.high[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m6"],
    -games_howell_result$conf.high[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m2"],
    games_howell_result$conf.low[games_howell_result$group1 == "m2"   & games_howell_result$group2 == "m6"]
  ),
  conf.high = c(
    games_howell_result$conf.low[games_howell_result$group1 == "m16t" & games_howell_result$group2 == "m2"],
    games_howell_result$conf.low[games_howell_result$group1 == "m16t" & games_howell_result$group2 == "m6"],
    -games_howell_result$conf.low[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m16t"],
    -games_howell_result$conf.low[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m6"],
    -games_howell_result$conf.low[games_howell_result$group1 == "m16"  & games_howell_result$group2 == "m2"],
    games_howell_result$conf.high[games_howell_result$group1 == "m2"  & games_howell_result$group2 == "m6"]
  )
)

ggdat$comparison = factor(ggdat$comparison, levels = ggdat$comparison)
```

```
ggplot(ggdat, aes(x = estimate, y = comparison)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0, linewidth = 0.8) +
  geom_point(size = 4) +
  theme_bw(base_size = 14) +
  theme(panel.grid.minor = element_blank()) +
  labs(
    x = "Estimated Difference",
    y = "Comparison"
  )
```

Figure 4: Pairwise Games-Howell CIs



3.5 Part e

Write a short paragraph directly addressing the researcher's hypothesis using your work in (a)-(d).

Solution

Games-Howell Pairwise comparisons among the untreated age groups demonstrated a statistically discernible increase in glucose hypermetabolism with age. SUV_{max} were significantly higher at 16 months compared to 2 months ($M_{diff} = 6.21, p = .007$, 95% CI [3.81, 8.60]), at 16 months compared to 6 months ($M_{diff} = 3.47, p = .008$, 95% CI [1.56, 5.39]), and at 6 months compared to 2 months ($M_{diff} = 2.73, p = .012$, 95% CI [1.31, 4.15]).

Low-dose nicotine treatment at 16 months significantly reduced glucose hypermetabolism, compared to the 16-month-old control group ($M_{diff} = 4.38, p = .043$, 95% CI [0.24, 8.51]). Notably, SUV_{max} in the treatment group did not differ significantly from those of the younger, untreated 2-month group ($M_{diff} = 1.83, p = .308$, 95% CI [-3.30, 6.96]) or the 6-month group ($M_{diff} = -0.90, p = .698$, 95% CI [-5.47, 3.67]).

References

Bracht, Thilo, Cornelia Mölleken, Maike Ahrens, et al. 2016. "Evaluation of the Biomarker Candidate MFAP4 for Non-Invasive Assessment of Hepatic Fibrosis in Hepatitis C Patients." *Journal of Translational Medicine* 14 (1): 201. https://doi.org/10.1186/s12967-016-0952-3.

Yang, Liang, Junfeng Shen, Chunhua Liu, et al. 2023. "Nicotine Rebalances NAD+ Homeostasis and Improves Aging-Related Symptoms in Male Mice by Enhancing NAMPT Activity." *Nature Communications* 14 (1): 900.